Note

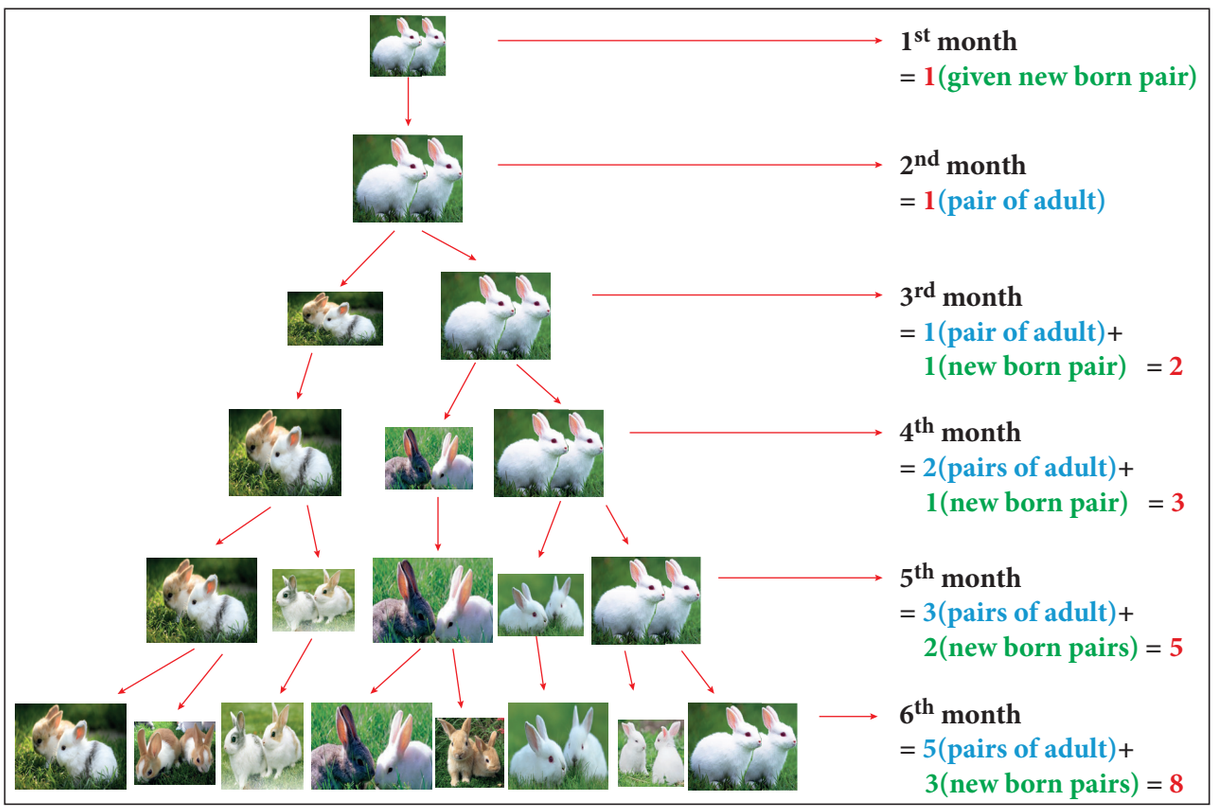
The difference between two consecutive numbers of the Fibonacci sequence increase very quickly. ((For example F(5) – F(4) = 5 – 3 = 2 ; F(10) – F(9) = 55 – 34 = 21 ; F(15) – F(14) = 610 – 377 = 233)(1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144, 233, 377, 610, 987, 1597, 2584, 4181, 6765, 10946 ...))

Example 7.7

Given that one pair of new born rabbits they produce a new pair each month and from the second month, each new pair can breed themselves. Find how many pairs of rabbits are bred from one pair in a year, and find the relationship between the number of months and the number of pairs of rabbits by tabulation (a pair means (a male and a female)).

Solution:

The below picture clearly forms the sequence is **1,1,2,3,5,8...** Here, we find the pattern in which each number is in the Fibonacci sequence, obtained by adding together with previous two. Going on like this to find subsequent numbers at the **twelfth month,** we will get **144 pairs of rabbits.** In the other words, twelfth Fibonacci number is 144.



Number of months	1	2	3	4	5	6	7	8	9	10	11	12
Number of pairs of rabbits	1	1	2	3	5	8	13	21	34	55	89	144

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